

Chapter

Arrays

1. Which of the following statements is correct for initializing a String as “A robot is a programmable machine”? MAIN 2024

- (a) `char x[ ] = {A, r, o, b, o, t, i, s, a, p, r, o, g, r, a, m, m, a, b, l, e, m, a, c, h, i, n, e};`
- (b) `String x = “A robot is a programmable machine”;`
- (c) `StringBuffer x = new StringBuffer(“A robot is a programmable machine”);`
- (d) All of the above.

2. A single-dimensional array contains N elements. What will be the last subscript?

- (a) N
- (b) N-1
- (c) N-2
- (d) N+1

3. What is the key difference between linear search and binary search? MAIN 2024

- (a) Linear search is faster than binary search.
- (b) Binary search works only on sorted arrays, while linear search works on both sorted and unsorted arrays.
- (c) Linear search divides the array into two halves, while binary search scans each element.
- (d) Binary search requires more time than linear search.

4. Which of the following is true about the .length property in Java? SQP 2023

- (a) It can only be used with single-dimensional arrays.
- (b) It returns the total memory size of the array.

(c) It returns the number of elements in a onedimensional array or the size of a row in a two-dimensional array.

(d) It always returns the size of the array minus one.

5. In the declaration `int x [ ] [ ] = new int [3][50];`, what does `x [0]. length` return? MAIN 2022

(a) 3

(b) 50

(c) 150

(d) Undefined

6. A programmer writes the following code to search for a value in an array using binary search:

```
if (value > mid) {  
    search = "First Half";  
}
```

This is incorrect. What will be the correct statement?

A. `if (value < mid) {search = "First Half";}`

B. `if (value > mid) {search = "Second Half";}`

C. `if (value == mid) {search = "Found";}`

D. All of the above

(a) Only A

(b) Both A and B

(c) Both B and C

(d) All of the above.

7. State the total size in bytes of the arrays `a[4]` of char data type and `p[4]` of float data type. MAIN 2024

8. Write a Java statement for the following purposes: COMP 2022

(i) Declare and create an array NUM of 40 integers.

(ii) Declare and create an array STR of 20 names.

(iii) Initialize an array of integers to store numbers -10, -8, -6, ..., 12.

(iv) Initialize an array of 5 names of convent schools:

ST. MARY, ST. THOMAS, ST. JOSEPHS ACADEMY, ST. PAULS ACADEMY, and ST. PETERS ACADEMY.

(v) Create an array of 10 real numbers named as REAL.

9. Write the statements to: COMP 2022

(i) Declare an array of strings with 3 rows and 5 columns.

(ii) Input the values of this array from the user in row-wise order.

(iii) Input values from the user in column-wise Order

10. How many locations are reserved for the following arrays? SQP 2023

(i) `int x[ ] = new int[25];`

(ii) `float a[ ] = new float[42];`

11. Define a class to declare an array to accept and store ten words. Display only those words which begin with the letter A or a and also end with the letter A or a. SQP 2022

EXAMPLE:

Input: Hari, Anita, Akash, Amrita, Alina, Devi, Rishab, John, Farha, AMITHA

Output:

Anita

Alina

AMITHA

12. Write a program to input 10 integer elements in an array and sort them in descending order using the bubble sort technique. SQP 2024

13. Define a class and store the given city names in a single-dimensional array.

Sort these names in alphabetical order using the Bubble Sort technique only.

INPUT Delhi, Bangalore, Agra, Mumbai, Kolkata **OUTPUT** :Agra, Bangalore,

Delhi, Kolkata, Mumbai **COMP 2023**

14. Write a program to accept a list of 50 integers from the user and another value to search in the list. Your program should print the position of the value in the list. COMP 2022

15. Accept a matrix of 5×5 from the user and check whether it is an identity matrix or not. An identity matrix is one whose all diagonal elements are 1 and all other elements are zero. COMP 2022

Solution

1. (d) All of the above

All the given options are valid in Java:

- (a) uses a char array to store individual characters.
- (b) uses a String, which is immutable.
- (c) uses a StringBuffer, which is mutable.

2. (b) N-1

In a single-dimensional array, indexing starts from 0. So, for N elements, the last element's subscript is N-1.

Example: `int[] arr = {10, 20, 30, 40};` → last element `arr[3]` ($4-1 = 3$).

3. (b) Binary search works only on sorted arrays, while linear search works on both sorted and unsorted arrays.

- **Linear Search:** Scans each element sequentially; works on sorted or unsorted arrays; time complexity $O(n)$.
- **Binary Search:** Requires a sorted array; repeatedly divides the array in half to locate the element; time complexity $O(\log n)$, making it faster for large datasets.

4. (c) It returns the number of elements in a one-dimensional array or the size of a row in a two-dimensional array.

In Java, the `.length` property of an array gives the total number of elements in that array. For a one-dimensional array, it gives the total elements, and for a two-dimensional array, using `.length` on the array gives the number of rows, while `.length` on a specific row gives the number of columns in that row.

5. (b) 50

In the declaration `int x[][] = new int[3][50];`, a 2D array with 3 rows and 50 columns is created. `x[0].length` returns the number of elements in the first row, which is 50.

6. (d) All of the above

In binary search, if the value is less than mid, it lies in the first half; if it is greater, it lies in the second half. If the value equals mid, it is found. Statements A, B, and C together correctly handle all these conditions.

7. The total size of the arrays is:

For `a[4]` (char type): Each char takes 2 bytes, so $4 \times 2 = 8$ bytes.

For `p[4]` (float type): Each float takes 4 bytes, so $4 \times 4 = 16$ bytes.

8. (i) `int NUM[] = new int[40];`

(ii) `String STR[] = new String[20];`

(iii) `int nums[] = {-10, -8, -6, -4, -2, 0, 2, 4, 6, 8, 10, 12};`

(iv) `String schools[] = {"ST. MARY", "ST. THOMAS", "ST. JOSEPHS ACADEMY", "ST. PAULS ACADEMY", "ST. PETERS ACADEMY"};`

(v) `float REAL[] = new float[10];`

9.

```
(i) String str[][] = new String[3]
    [5];
(ii) Scanner sc = new Scanner(System.
    in);
    System.out.println("Input
    values into the array in row-wise
    order:");
    for (int i = 0; i <= 2; i++) { //
Loop for Row
        for (int j = 0; j <= 4; j++) {
            // Loop for Column
            str[i][j] = sc.next();
        }
    }
(iii) Scanner sc = new Scanner(System.
    in);
    System.out.print("Input the
    values into the 2-dimension array");
    for (int i = 0; i <= 4; i++) { //
Loop for Column
        for (int j = 0; j <= 2; j++) {
            // Loop for Row
            str[j][i] = sc.next();
        }
    }
```

10. (i) 25

(ii) 42

11.

```
import java.util.Scanner;

public class FilterWords {
    public static void main(String[]
args) {
        Scanner sc = new Scanner(System.
in);
        String words[] = new String[10];
        System.out.println("Enter 10
words:");
        for (int i = 0; i < 10; i++) {
            words[i] = sc.nextLine();
        }
        System.out.println("Words
beginning and ending with A or a:");
        for (int i = 0; i < 10; i++) {
            String word = words[i];
            if ((word.startsWith("A")
|| word.startsWith("a")) &&
                (word.endsWith("A")
|| word.endsWith("a"))) {
                System.out.println(word);
            }
        }
    }
}
```

12.

```
import java.util.Scanner;

public class BubbleSort {
    public static void main(String[]
args) {
        int a[] = new int[10];
        Scanner scan = new Scanner(System.
in);
        System.out.println("Enter 10
integers:");
        for (int i = 0; i < 10; i++) {
            a[i] = scan.nextInt();
        }
    }
}
```

```

        // Bubble sort in descending
order
        for (int i = 0; i < 9; i++) {
            for (int j = 0; j < 9 -
i; j++) {
                if (a[j] < a[j + 1]) {
                    int temp = a[j];
                    a[j] = a[j + 1];
                    a[j + 1] = temp;
                }
            }
        }
        System.out.println("Sorted
Array in Descending Order:");
        for (int i = 0; i < 10; i++) {
            System.out.println(a[i]);
        }
    }
}

```

Variables Used:

a: Array of integer elements

k, i, j: Loop variables

t: Temporary variable used in the sorting process
for swapping elements

13.

```

class sortArray {
    public static void main() {
        String city[] = {"Delhi",
"Bangalore",    "Agra",    "Mumbai",
"Kolkata"};
        String temp;
        int i, j;
        int n = 5;

        for (i = 1; i <= n - 1; i++) {
            for (j = 0; j <= n - i -
1; j++) {
                if ((city[j].
compareTo(city[j + 1])) > 0) {
                    temp = city[j];
                    city[j] = city[j
+ 1];
                    city[j + 1] = temp;
                }
            }
        }
    }
}

```

```

        System.out.println("Cities in
alphabetical order:");
        for (i = 0; i < n; i++) {
            System.out.println(city[i]);
        }
    }
}

```

Variables Used:

city[]: Stores list of cities

i, j, k: Loop variables

temp: Temporary string variable

14.

```

import java.util.*;
class Q22 {
    public static void main(String[]
args) {
        Scanner obj = new Scanner(System.
in);

        int pos = -1;
        System.out.println("Enter
the value whose position is to be
searched:");
        int v = obj.nextInt();
        int a[] = new int[50];
        for (int i = 0; i < 50; i++) {
            System.out.println("Enter
the number:");
            a[i] = obj.nextInt();
        }
        for (int i = 0; i < 50; i++) {
            if (v == a[i]) {
                pos = i + 1;
                break;
            }
        }
        if (pos == -1) {
            System.out.println("Value
not found in the list.");
        } else {
            System.out.println("Position
= " + pos);
        }
    }
}

```

15.

```
import java.util.*;
class {
    public static void main(String[]
args) {
    Scanner obj = new Scanner(System.
in);

    int n[][] = new int[5][5];

    System.out.println("Enter the
matrix elements");
    for (int i = 0; i < 5; i++) {
        System.out.println("Enter
the no of " + i + "th row");
        for (int j = 0; j < 5;
j++) {
            n[i][j] = obj.
nextInt();
        }
    }

    System.out.println();
    boolean v1 = false;
    boolean v2 = false;

    for (int i = 0; i < 5; i++) {
        for (int j = 0; j < 5;
j++) {
            if (i == j) {
                if (n[i][j] == 1)
                    v1 = true;
            } else if (n[i][j]
== 0) {
                v2 = true;
            }
        }
    }

    if (v1 == true && v2 == true)
        System.out.println("Identity
matrix");
    else
        System.out.println("Not
identity");
    }
}
```

```

        int x[][] = {{6, 9, 0, 8},
{3, 2, 1, 5}, {2, 9, 0, 1}, {4, 1,
9, 6}};
        System.out.println("1. Sum of
main diagonal elements");
        System.out.println("2. Sum of
right diagonal elements");
        char c = buf.next().charAt(0);
        switch (c) {
            case 1:
                int s = 0;
                for (int i = 0; i <=
3; i++) {
                    s = s + x[i][i];
                }
                System.out.println(s);
                break;
            case 2:
                int r = 0;
                int j = 3;
                for (int i = 0; i <=
3; i++) {
                    r = r + x[i][j];
                    j--;
                }
                System.out.println(r);
                break;
            default: System.out.println("\nWrong
choice entered");
        }
    }
}

```